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Full length article

Climate- and technology-specific PUE and WUE estimations for U.S. data centers using a hybrid statistical and thermodynamics-based approach

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ABSTRACT

The water use of data centers (DCs) is becoming an increasingly important consideration for sustainability analysts, but has been difficult to quantify due to lack of reported data by DC operators. This work develops thermodynamically-compatible water use effectiveness (WUE) and power usage effectiveness (PUE) ranges needed for direct and indirect water use analysis for 10 different DC archetypes in 15 U.S. climate zones, using a hybrid physical-statistical approach that is validated with real-world data. The ranges further capture variabilities between best and poor DC efficiency practices, showing that PUE and WUE values can have relative difference as much as 62% and 100% respectively, depending on cooling technologies, efficiencies, and locations. PUE results indicate significant climate effects, whereas WUE results indicate the strongest climate effects for airside economizers with either adiabatic or water-cooled chiller systems. Collectively, the results underscore the importance of considering technology, efficiency, and climate effects when estimating PUE and WUE for direct and indirect DC water use analysis. Results also identify variables leading to best-achievable PUE and WUE values by climate zone and cooling system type—including operational set points, use of free cooling, and cooling tower equipment and operational factors—which can support DC water- and energy-efficiency policy initiatives.

1. Introduction

Data centers (DCs) are an increasingly important component of global electricity use. This electricity use may grow in the near-term due to rising demand for DC services as the world becomes increasingly interconnected and as compute-intensive applications like artificial intelligence and blockchains become commonplace (Lei et al., 2021b; Masanet et al., 2020a; Patterson et al., 2021). As a result, the electricity use of DCs has become a frequent topic of research within the sustainability analysis community (Hintemann and Hinterholzer, 2019; Shehabi et al., 2018). DC electricity use can generally be divided into the electricity use of: (1) information technology (IT) equipment (i.e., servers, storage devices, and network devices) and (2) power provision and space conditioning (i.e., infrastructure) equipment (Lei, 2020; Masanet et al., 2020a; Shehabi et al., 2016).

Many analysts model infrastructure equipment energy use by assuming average power usage effectiveness (PUE) values, where PUE is defined as the ratio of a DC's total electricity use (in kilowatt-hours, or kWh) to its IT electricity use (in kWh) (Jaureguialzo, 2011). The analyst's assumed PUE value is then multiplied by modeled IT electricity use to arrive at a total electricity use estimate for a DC or for DCs within a given region (European Commission, 2020; Koomey, 2011; Shehabi et al., 2016). Total electricity use can also be multiplied by water consumption factors (WCFs) corresponding to local electricity sources to estimate the indirect water use associated with DC operations (Peer et al., 2019; Siddik et al., 2021).

In addition to indirect water use, the onsite water use of DCs is also an important environmental consideration (Mytton, 2021; Siddik et al., 2021), and can even pose barriers to DC siting in water stressed regions. Indeed, some DC facilities have been scrutinized by local communities, public utilities, and water conservation groups for requiring too much water (SATTIRAJU, 2020). Onsite water use is driven by DC space conditioning systems, which are comprised of cooling systems and air humidification systems (Lei and Masanet, 2020). The major types of cooling systems that require onsite water are water-cooled chiller systems and direct evaporative cooling systems (Evans, 2004). One metric

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for onsite water use is water usage effectiveness (WUE), which is defined as the ratio of a DC's total onsite water use to its IT electricity use (Patterson et al., 2011), expressed in units of liters per kWh. Despite the growing importance of onsite water use, few DC operators regularly report their WUE values, with Facebook (Facebook, 2014) and Scaleway (Scaleway, 2021) being notable exceptions.

To incorporate water use estimation into DC sustainability models, analysts require credible estimates of both PUE and WUE values that correspond not only to specific cooling system types, but also to specific local climate conditions. Local climate conditions can be important factors because they affect a DC's ability to utilize economizers for "free cooling" in lieu of mechanical cooling, while local air temperatures and relative humidity (RH) will affect space conditioning requirements (Lei and Masanet, 2020). Moreover, because WUE and PUE values are interrelated on the basis of mass and energy balances that are governed by thermodynamic principles, for consistent modeling of both onsite and indirect water use, analysts must be careful to choose WUE values that are thermodynamically-compatible with their assumed PUE values. To date, however, no thermodynamically-compatible WUE and PUE values have been published for use by sustainability analysts across a range of typical DC space conditioning types and climatic conditions. Therefore, the analysis community has been limited in its ability to accurately estimate DC onsite and indirect water use.

This paper helps overcome this analysis barrier by presenting a hybrid physical-statistical approach for predicting thermodynamically-compatible WUE and PUE ranges with reasonable accuracy for different DC types and locations, and which only rely on publicly-available data and reasonable and transparent assumptions about key system parameters.

More specifically, this paper makes the following contributions to the literature:

- Expands a previous and proven physics-based DC PUE model (Lei and Masanet, 2020) to generate thermodynamically-compatible WUE estimates.
- Constructs 10 different DC archetypes on the basis of detailed technology and operating parameters, which broadly represent most types of air-cooled DC operations in the United States.
- Applies the model to generate predicted PUE and WUE ranges (bounded by best and poor practices) for each DC archetype within 15 different U.S. climate zones, which can be used by the sustainability analysis community for estimating water use under many different conditions and locations.
- Presents and discusses the model's sensitive variables, which are those that future analysts should focus on for improving the accuracy of the model when applied to any specific DC.
- Discusses the main climate, technological, and operational determinants of PUE and WUE and the key factors required for reaching best-practice values.

Furthermore, the model is made freely available for use and refinement by the research community. This paper does not quantify the absolute onsite or indirect water use of DCs; however, the WUE and PUE ranges presented here can be used for such purposes in future studies provided they quantify IT loads and select WCFs for specific DC locations.

2. Previous data center water use and WUE analyses

Numerous previous papers have proposed physics-based PUE models and/or presented PUE data (see (Lei and Masanet, 2020) for a comprehensive review). In contrast, only a few studies of data center direct water use exist in the peer-reviewed literature, as summarized in Table 1. These studies fall into two types that are relevant to DC water use modeling: (1) those that quantified the water use of different cooling systems, and (2) those that quantified or proposed WUE values

Table 1
Summary of previous studies of DC water use.

Study	Study scope (direct and/or indirect?)	Direct water scope (cooling tower, adiabatic cooling, space humidification?)	Were PUE values reported?	Were WUE values reported?	DC locations considered?	Cooling variants considered?	Study type (empirical data or modeled estimate)	Temporal basis (single day, multiple day, annual)	Level of technical detail reported (equipment efficiency, setpoints, etc.)	If a model was used, is it available and generalizable?
(Sharma et al., 2009, Sharma et al., 2008)	Direct	Cooling tower	No	No	No	No	Empirical data	Multiple day	No	Not used
(Sharma et al., 2010)	Direct + indirect	Cooling tower	No	No	Palo Alto, California	Yes	Empirical data	Multiple day	No	Not used
(Shehabi et al., 2016)	Direct + indirect	Cooling tower	Yes	No	United States	n/a	Empirical data	Annual	No	Not used
(Sharma et al., 2017)	Direct	Cooling tower	Yes	Yes	Holyoke, Massachusetts	No	Empirical data	Annual	Supply air temperature (26.7 °C)	Not used
(Gozcu et al., 2017)	Direct	Cooling tower + adiabatic cooling	Yes	Yes	19 cities	Yes	Modeled estimate	Annual	Chiller efficiency (COP=4.0), Supply air temperature (15 or 25 °C)	Yes (Not publicly available and not fully generalizable)
(Ristic et al., 2015)	Direct + indirect	Cooling tower + adiabatic cooling	No	No	Phoenix, Arizona	Yes	Empirical data	Annual	Supply air temperature (27°C)	Not used
(Siddik et al., 2021)	Direct + indirect	Cooling tower	Yes	No	United States	n/a	Empirical data	Annual	No	Not used

Note: n/a = not applicable.

corresponding to specific conditions.

Studies of the first type enable sustainability analysts to understand the drivers of DC water and energy use in different cooling systems and locations as well as the potential magnitudes of absolute water use at different scales. The earliest such studies were published by Sharma et al., who measured water and energy data from a single DC cooled by a hybrid system employing air-cooled and water-cooled chillers operating on a rotational basis. The authors analyzed two representative days to evaluate the tradeoffs between water use and energy efficiency, concluding that water-cooled chillers are more energy efficient than air-cooled chillers but that the former drives significant cooling tower water consumption (Sharma et al., 2009, 2008). The same authors later analyzed the use of an airside economizer, and compared its water and energy use to a water-cooled chiller system and an air-cooled chiller system using performance estimations based on manufacturer data (Sharma et al., 2010). They found that compared with using a water-cooled chiller, using an air-cooled chiller can reduce the DC water consumption at the expense of increasing electricity use, while using outside air cooling could reduce both water and electricity use.

Other studies of this type focused on estimating aggregate quantities of water use associated with DC cooling at different scales. Ristic et al. (Ristic et al., 2015) estimated a global DC water footprint range based on the direct water use of four different cooling technologies in Phoenix, a worldwide DC energy use estimate (Koomey, 2011), and a global average electricity WCF, resulting in a large range of 1047 to 151,061 m^3 per TJ of DC energy use. A 2016 report by Lawrence Berkeley National Laboratory (LBNL) estimated the total direct water use of DCs at the national level, estimating a quantity of 126 billion liters in 2014 (Shehabi et al., 2016). To arrive at this estimate, the authors assumed a nationwide average DC on-site water use intensity of 1.8 liters per kWh of total DC site energy use. Siddik et al. investigated the water footprint of U.S. DCs in 2018 (Siddik et al., 2021), based on the LBNL onsite water use intensity (Shehabi et al., 2016), average DC PUE values reported in (Masanet et al., 2020b), and an estimated geographical distribution of U.S. DCs (Ganeshalingam et al., 2017), concluding that around 1/5 of the U.S. DC servers are cooled by water drained from water stressed watersheds.

Only two studies of the second type have been published. Sharma et al. performed a time-series analysis of the energy and direct water use of a DC employing a water-cooled chiller system with a waterside economizer in western Massachusetts using monitored data (Sharma et al., 2017). The authors observed clear seasonal variations in the DC's on-site PUE and WUE values (both of which peaked during the summer season), highlighting the significance of climate effects on the DCs' power and water use. A study by Gozcu et al. attempted to model DC PUE and WUE values, further analyzing the energy and direct water saving potentials of different DC free cooling technologies (including airside economizers, indirect evaporative coolers, waterside economizers) in 19 cities, underscoring the importance of economizer choices in DC operations (Gozcu et al., 2017). However, the modeling methodology of this study was not clearly documented, and the modeling results were not verified using real DC reported data.

While all of these studies provided valuable insights into the water use of data centers at different spatial and temporal scales, none answer our proposed research questions due to several important limitations. First, onsite water use estimates are typically limited to cooling tower water use only, which neglects potentially non-trivial water consumption by adiabatic cooling and humidification systems in many DCs. Second, onsite water use estimates have mostly been limited to a small set of cooling technologies in a few specific locations without revealing important technical parameters that may impact WUE in those locations, such as equipment efficiencies and set points. Third, all but one study (Gozcu et al., 2017) analyzed the direct water use of single DCs in specific locations, and using data from a small number of representative days, which prohibits exploration of how climate variations by location would affect WUE. Even so, the study by Gozcu et al. applied static equipment power and water use profiles and did not explore how DC facility system variations would affect WUE. Fourth, while some studies presented corresponding PUE values, others did not, which overlooks an important consideration for estimating both direct and indirect water use intensities in a thermodynamically-consistent manner. Finally, the overall lack of generalizable models prohibits extending methods and results to other cooling system types, operating conditions, or locations, which is necessary for sustainability analysts and to estimate DC water use and identify improvement pathways under the many different conditions associated with existing DCs.

3. Methodology

Fig. 1 illustrates the methodological approach employed in this study, which is further described in the sections that follow.

3.1. Modeling scope and system definitions

In this paper we focus on modeling the WUE and PUE of DCs under different space conditioning technology, operations, and climate conditions. We report ranges of both WUE and PUE values corresponding to best and poor efficiency practice limits, where PUE values can be useful in future studies for estimating indirect water use. Importantly, WUE and PUE are directly related by physics, so it is important that compatible values are used together in future studies to avoid introducing errors that can be made when predicting WUE and PUE independently.

A DC's space conditioning technologies, operating characteristics, and climate conditions all influence both its PUE and WUE values. In our analysis, we selected seven different space conditioning technology configurations consisting of different combinations of refrigeration units, cooling towers, direct evaporative (i.e., adiabatic) coolers, and space humidification options as summarized in Table 2. These configurations are most representative of present-day DC space conditioning systems globally based on (Capozzoli and Primiceri, 2015; Evans, 2012; Joshi and Kumar, 2013; Lei and Masanet, 2020) and are schematically depicted in Fig. 2. For further information and technical details on these systems, readers are referred to (Evans, 2012; Lei and Masanet, 2020).

To assess differences in operating characteristics that affect WUE and PUE, including equipment efficiencies and internal temperature and RH set points, we further defined three different DC size classes based on the taxonomy in (Shehabi et al., 2016). Small DCs are typically less than 1000 ft^2, which generally operate with low uninterruptible power supply (UPS) and airflow efficiencies and narrow allowable temperature and RH ranges (Ni and Bai, 2017). Large-scale DCs occupy more than 20,000 ft^2, inclusive of hyperscale DCs, and tend to operate with high UPS and airflow efficiencies with wider allowable temperature and RH ranges, often leading to frequent use of free cooling (Barroso et al., 2013; Miller, 2009; Park, 2011). In between these two extremes lie midsize DCs, which are representative of many on-premises enterprise and colocation DCs ranging from 1000 to 20,000 ft^2 and with average operating efficiencies (Lawrence et al., 2019).

In total, we considered 10 different cases representing different size classes and space conditioning technology configurations, covering a wide swath of typical DC technology and operating conditions.

Finally, to consider how local climate conditions affect WUE and PUE values, we focused on the United States given its wide variety of climate zones. Specifically, we analyzed each DC case within fifteen distinct U.S. climate zones designated by the International Energy Conservation Code (IECC) and the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) (ASHRAE, 2013; Baechler et al., 2015). The IECC/ASHRAE climate zones include eight major temperature-oriented zones and three moisture subcategories (A: moist; B: dry; C: marine.) as depicted in Fig. 3 (figure produced using R 'usmap' package (Lakicevic, Povak and Reynolds, 2020)). For each IECC/ASHRAE climate zone, we chose a representative city designated by the U.S.

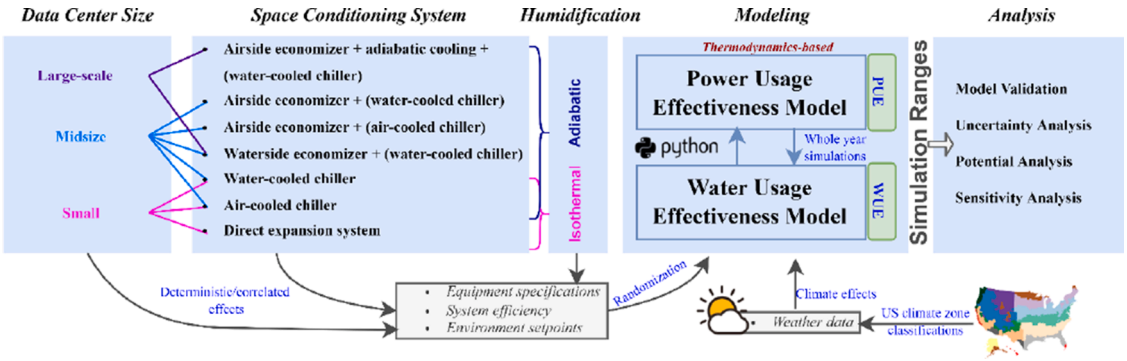


Fig. 1. Simplified diagram of the research methodology.

Table 2
DC size class and space conditioning cases considered.

Case	DC Size	Cooling system configuration (1)	Direct water use equipment included		
			Direct evaporation	Space humidification (2)	Cooling tower
1	Large-scale	Airside economizer + adiabatic cooling + (water-cooled chiller)	✓	✓ (adiabatic)	✓
2	Large-scale	Waterside economizer + (water-cooled chiller)	n/a	✓ (adiabatic)	✓
3	Midsize	Airside economizer + (water-cooled chiller)	n/a	✓ (adiabatic)	✓
4	Midsize	Waterside economizer + (water-cooled chiller)	n/a	✓ (adiabatic)	✓
5	Midsize	Water-cooled chiller	n/a	✓ (adiabatic)	✓
6	Midsize	Airside economizer + (air-cooled chiller)	n/a	✓ (adiabatic)	n/a
7	Midsize	Air-cooled chiller	n/a	✓ (adiabatic)	n/a
8	Small	Water-cooled chiller	n/a	✓ (isothermal)	✓
9	Small	Air-cooled chiller	n/a	✓ (isothermal)	n/a
10	Small	Direct expansion system	n/a	✓ (isothermal)	n/a

Note: (1) parentheses indicate a supplemental cooling system (e.g., in case 1, a water-cooled chiller will be used only when the airside economizer and adiabatic cooling system cannot satisfy the DC cooling requirement); (2) parentheses indicate the type of humidifiers being used for space humidification (i.e., adiabatic humidifier or isothermal humidifier); ✓ = yes; n/a = not applicable.

Department of Energy (Department of Energy, 2020), shown in brackets in Fig. 3. For each city, meteorological data containing hourly air temperature, RH, and atmospheric pressure in a typical meteorological year were compiled (EnergyPlus, 2016), which were then used to simulate WUE and PUE values for each DC case for each hour over the entire meteorological year.

In the next section, we briefly introduce the hourly DC PUE and WUE model used in our framework. Our DC PUE and WUE simulation code (in Python) is available through the link provided in Appendix A.

3.2. DC PUE and WUE model

Our combined PUE and WUE model is derived from a physics-based PUE model previously published in (Lei and Masanet, 2020). Below, we describe the governing equations and key variables in the model for calculating both WUE and PUE for a given set of space conditioning system technology, operating, and climate conditions. For a more detailed description of the model and calculation approach, readers are referred to Appendix A and (Lei and Masanet, 2020). The numerical values assigned to each variable in our DC case simulations are described in the next section.

Eq (1) expresses the DC on-site water use rate (w^{DC}, kg/s) which is a function of the water used by three equipment types (see Table 2): (1) cooling towers (including evaporated water (w_e^{CT}, kg/s), windage loss of water (w_w^{CT}, kg/s), and draw-off water (w_d^{CT}, kg/s)); (2) adiabatic cooling (also called as direct evaporative cooling) (w^{AC}, kg/s); and (3) space humidification (w^{SH}, kg/s)).

$$w^{DC} = w^{CT} + w^{AC} + w^{SH} = \left(w_e^{CT} + w_w^{CT} + w_d^{CT}\right) + w^{AC} + w^{SH} \quad (1)$$

Cooling tower water use rates in Eq (1) depend on the amount of heat generated in the data center (Q^{DC}, kW) and the power use of the DC (p^{DC}, kW), which is calculated using Eqs. (2) and (3). These two equations are also fundamental for simultaneously calculating both WUE and PUE for the same system conditions on the basis of underlying system physics. Further details on the calculation of total DC power use in Eq (3) can be found in (Lei and Masanet, 2020).

$$Q^{DC} = p^{IT} + p_{loss}^{UPS} + p_{loss}^{PD} + p^{L} + p^{IS} \quad (2)$$

$$p^{DC} = p^{IT} + p_{loss}^{UPS} + p_{loss}^{PD} + p^{L} + p^{IS} + \sum p^{PUMPs} + \sum p^{FANs} + p^{CH} \quad (3)$$

where p^{IT}, p_{loss}^{UPS}, p_{loss}^{PD}, p^{L}, p^{IS}, p^{PUMPs}, p^{FANs}, and p^{CH} are respectively the power use of IT equipment, power loss of UPS system, power loss of power transformation and distribution system, and the power used by the lighting system, isothermal humidifier, pumps (which includes the humidification pumps used for adiabatic cooling and in adiabatic space humidifiers), fans, and mechanical chiller (direct expansion cooling units were classified into the chiller category in this paper for simplicity).

The energy flow across the cooling tower (Q^{CT}, kW) is calculated via Eq (4).

$$Q^{CT} = Q^{DC} + Q_s^{CH} / \left(SHR \times COP_{L,T}\right) \quad (4)$$

where Q_s^{CH} is the quantity of sensible cooling supplied by a mechanical chiller (kW), governed by the DC heat generation and the amount of free cooling supplied by the economizer. The amount of free cooling being exploited is a function of outdoor air conditions and indoor air setpoints, which is determined by the enthalpy difference between supply and outside air for case 1, the temperature difference between supply and

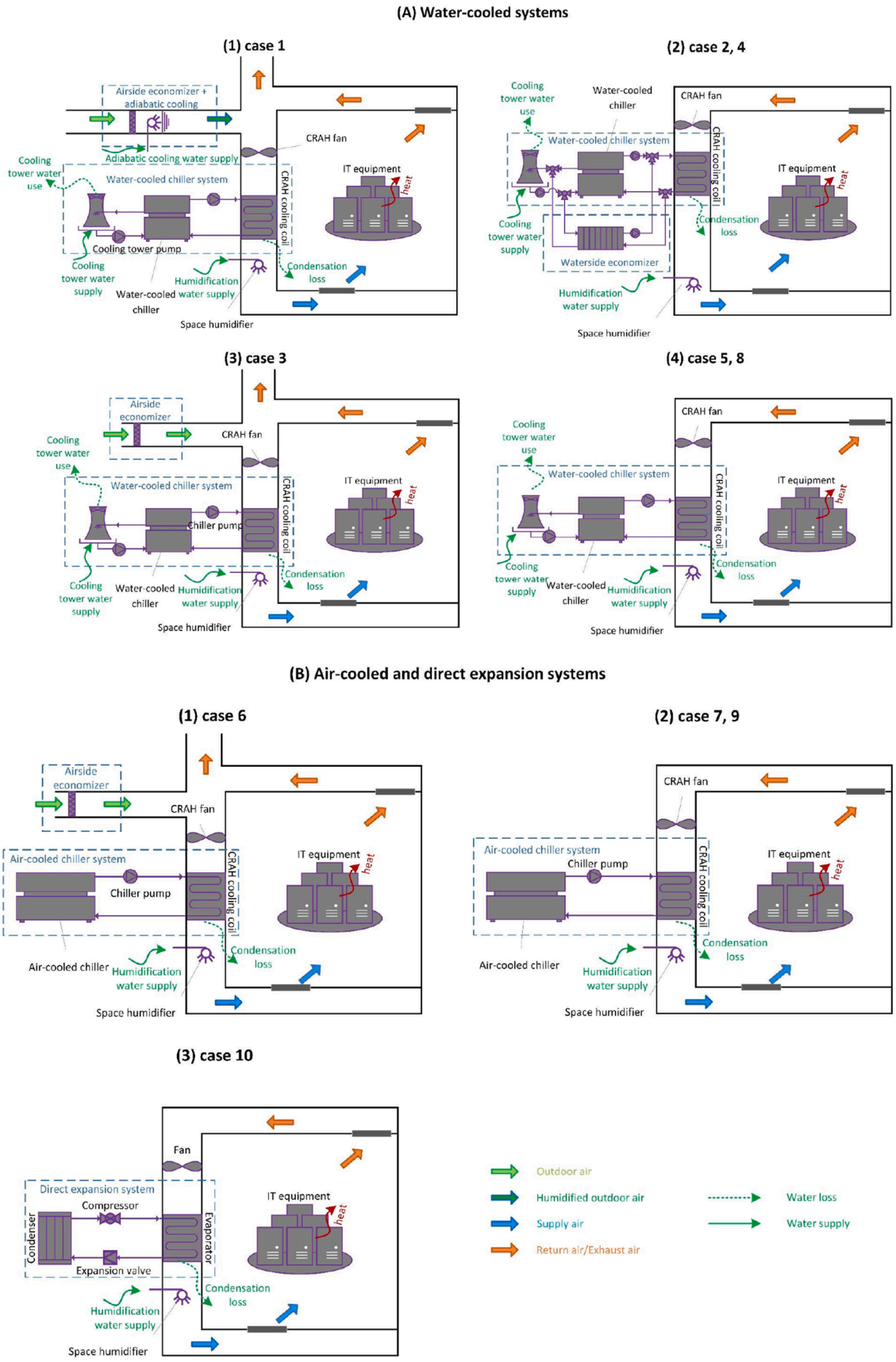


Fig. 2. Schematic diagrams of considered DC space conditioning system configurations (CRAH = computer room air handler).

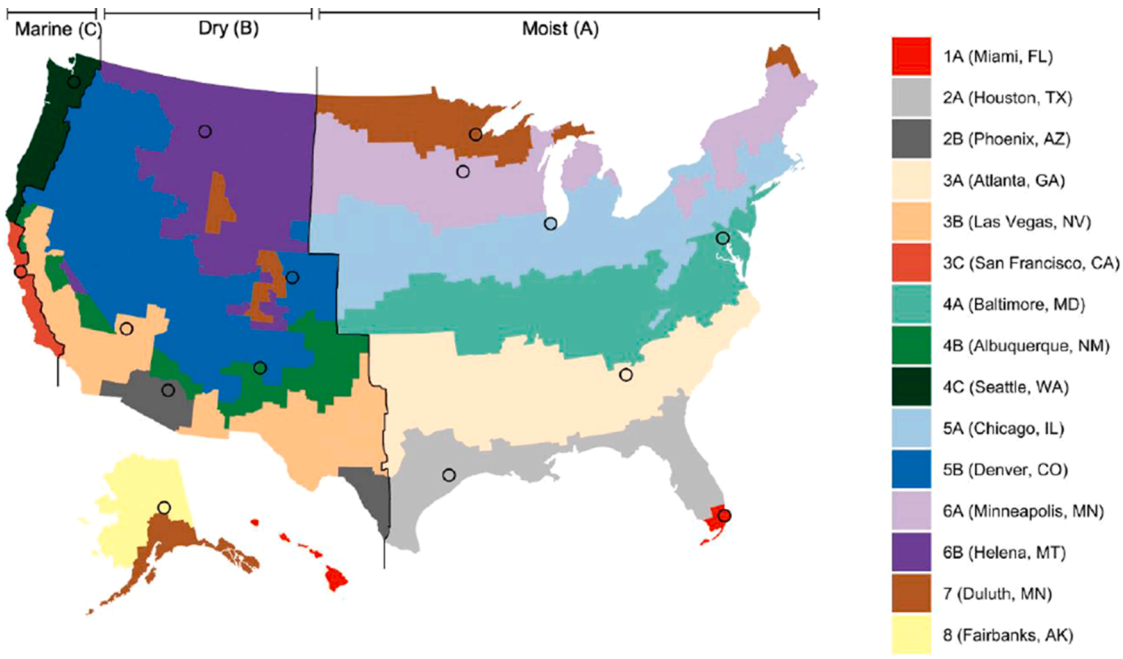


Fig. 3. United States IECC/ASHRAE climate zones (circles represent climate zone representative cities).

outside air for other cases employing an airside economizer, and the temperature difference between the return facility water temperature and the outdoor wet bulb temperature for cases employing a waterside economizer (Joshi and Kumar, 2013; Lei and Masanet, 2020). The variable *SHR* is the sensible heat ratio (kW/kW), defined as the sensible cooling load divided by the total cooling load of a mechanical chiller; $COP_{L,T}$ is the chiller's coefficient of performance (COP) ($kW\ /kW$), which is parametrized by the chiller load factor and an outdoor temperature-related variable using Gaussian process regression based on data from (Gullo et al., 2017; Squillo, 2018; Yu and Chan, 2007).

Thereafter, the three different cooling tower water use variables were calculated using an established cooling tower energy and mass balance model Environmental Protection Agency and program, 2017), expressed by Eqs. (5)-((7).

$$w_e^{CT} = Q^{CT} / H_{vap} \tag{5}$$

$$w_w^{CT} = \varnothing_w \times Q^{CT} / (c_w \times \Delta T_{CT}) \tag{6}$$

$$w_d^{CT} = w_e^{CT} / (CC - 1) - w_w^{CT} \tag{7}$$

where H_{vap} is the latent heat of vaporization of water ($kJ\ /kg$), which is a function of the cooling tower water temperature; $\varnothing_w$ is the windage loss of water as a percentage of the cooling tower mass flow rate of water; c_w is the specific heat of water ($kJ/kg{\cdot}^{\circ}C$); ΔT_{CT} is the cooling tower supply and return temperature difference (°C); and *CC* is the cycles of concentration, describing the levels of dissolved solids in the cooling tower water.

Our case 1 considers the use of airside economizers with adiabatic cooling, with reliance on a water-cooled chiller for hours when climate conditions are not favorable. This configuration can deliver both low WUE and low PUE values, especially in cooler climates where reliance on mechanical chillers can be substantially reduced or even eliminated completely as demonstrated by several hyperscale DC operators (Amazon, 2020; Morgan, 2016; Park, 2011). These operators also tend to use high temperature set points to maximize adiabatic cooling use hours (Miller, 2009; Park, 2011). The amount of water use for adiabatic cooling is determined by both outdoor air conditions and the indoor air set points (i.e., the dry bulb temperature, RH, and dew point) and is calculated using Eq. (8).

$$w^{AC} = m_{cd,\ dry} \times (d_{cd} - d_{oa}) \tag{8}$$

where $m_{cd,\ dry}$ is the mass flow rate of dry air that goes through the adiabatic process (kg/s); d_{cd} is the absolute humidity of humidified air (kg/kg) (see (Joshi and Kumar, 2013) for the calculation method); and d_{oa} is the absolute humidity of outdoor air (kg/kg).

Lastly, to compensate for humidity loss within the internal DC space and to avoid electrostatic discharge (ASHRAE, 2015), humidification of indoor air is occasionally required. Applying the law of mass conservation, the water use of space humidification is modeled as a function of the latent heat removed by the chiller system (Evans, 2008), which is expressed as Eq. (9). Note that the amount of water used for space humidification is not influenced by the choice of DC humidifier in the model (i.e., adiabatic humidifier or isothermal humidifier); however, this choice can result in a different PUE value given that these two humidification technologies have different effect on the supplied DC air temperature (Energy Star, 2020).

$$w^{SH} = \left(Q_s^{CH} / SHR - Q_s^{CH}\right) / H_{cond} \tag{9}$$

where H_{cond} is the latent heat of condensation of water (kJ/kg).

Consequently, the hourly PUE (PUE_i^{DC}, kWh/kWh) and hourly WUE (WUE_i^{DC}, kg/kWh or $liters/kWh$) are calculated in the model as follows:

$$PUE_i^{DC} = p^{DC} / p^{IT} \tag{10}$$

$$WUE_i^{DC} = 3600\ \times w^{DC} / p^{IT} \tag{11}$$

where the subscript *i* indicates the power or water use in the *i*th hour of a year.

Finally, the annual average PUE (PUE^{DC}, kWh/kWh) and WUE (WUE^{DC}, kg/kWh or $liters/kWh$) are calculated as follows:

$$PUE^{DC} = \sum_i PUE_i^{DC} \Big/ N \tag{12}$$

$$WUE^{DC} = \sum_i WUE_i^{DC} \Big/ N \tag{13}$$

where N is the total number of hours across a year (typically 8760 h).

3.3. Model inputs and data sources

To analyze a given DC case in a given climate zone, the combined PUE and WUE model requires two major categories of model inputs: (1) DC facility system variables (i.e., equipment specifications, system efficiency variables, and indoor environment setpoints); and (2) ambient climate data. Appendix B Table B.1 summarizes the ranges assumed for major facility system variables in each DC case, which were determined based on best available data sources and engineering specifications and estimations. Importantly, high and low values were established for each variable to facilitate model sensitivity and uncertainty analyses, with values corresponding to best-case and worst-case conditions for each parameter gleaned from the data sources and engineering estimation process. As such, resulting PUE and WUE ranges quantify the difference between best and poor efficiency practices for each case and climate zone. Hourly ambient climate data were collected for each representative city (see Fig. 3) based on meteorological data acquired from EnergyPlus Weather Data (EnergyPlus, 2016), which included hourly dry bulb temperature, RH, and atmospheric pressure in a typical meteorological year.

3.4. Uncertainty and sensitivity analysis

Parameter uncertainty is one of the major uncertainty sources in physics-based engineering models (Doukas et al., 2018). Because DCs are complex systems with considerable variations in design and system operation, simulation runs considering all possible DC facility system variable values are critical for understanding the uncertainties in the modeling results. In this study, the Latin hypercube sampling approach was applied for efficient parameter stratification, and for reducing simulation runs of the computationally-expensive Python model (McKay et al., 2000). For each climate zone and DC case, 50 samples of the DC facility system variables summarized in Appendix B Table B.1 were generated using Latin hypercube sampling, aiming at balancing the computation time and the convergence in the variance of the simulation results. Specifically, the sample size used in this study was selected based on computer experiments that quantify the stabilization in the simulation range result versus increased sample size, and we found that the resulting simulation range stabilized after passing 35 samples (see supporting information Fig. S2). Each of the generated samples represents one possible DC design and operation scenario and was used for the one-hour time-step PUE and WUE simulation across the year under the representative climates. As a result, the 50 one-year-round simulation results provide reasonable uncertainty ranges of the DCs' annual average PUE and WUE values for a given climate zone.

In addition, Sobol's sensitivity analysis was performed to identify and rank the model input variables that most contribute to variation in the simulated PUE and WUE values. Sobol's method has the advantage of decomposing the variance of the model's outputs to additive terms attributed to the model's input variables (Sobol, 2001). Two different sensitivity metrics of Sobol's method are typically employed by the sustainability modeling community to inspect complicated models: the main-effect index, and the total-effect index (Nossent et al., 2011). Given the high-level interactions of the model's input variables in affecting a DC's power and water use (e.g., climate conditions and DC indoor environment setpoints jointly determine its economizer use, hence its power and water use), the total-effect index was chosen here as the more robust sensitivity measure. A Monte Carlo estimator with Saltelli's sampling scheme was used in this study to numerically approximate the Sobol's total-effect index, and the associated sensitivity analysis computer experiments were conducted using the Python's SALib package (Herman and Usher, 2017). The detailed sensitivity analysis results are presented in Section 4.4.

4. Results and discussion

4.1. Model validation

To validate that the model produced accurate results for its intended purpose—i.e., providing climate- and technology-specific best-to-poor-practice ranges of PUE and WUE values for use by sustainability analysts—the simulation ranges were compared to reported values from specific DC operators (Facebook, 2020a; 2020b, 2020c, 2020d; Sharma et al., 2017). Fig. 4 shows the validation results, in which the technology case and climate zone for each reporting DC are indicated on the y-axis. All reported annual average PUE and WUE values lie within the ranges estimated by the model. Interestingly, most of the reported values lie toward the best practice end of the simulation ranges, which may indicate that energy and water efficient DCs are more likely to report their PUE and WUE values publicly. In addition, reported values for large-scale DCs with adiabatic cooling systems indicate that low PUE values may have been achieved at the expense of higher WUE values. The reason is that adiabatic humidification sprays water into air (resulting in a larger WUE values) in return for decreased supply air temperatures and hence lower PUE values. For midsize DCs, the estimated ranges are much larger compared to those of the large-scale DCs due to the wider variabilities in facility system variables and the nature of the cooling technologies chosen by large-scale DCs (see section 4.2). Finally, it can be observed that model accurately captures the decreasing trends of PUE and WUE ranges when moving toward more efficient large-scale DCs and cooler climates, which is further discussed in the next section.

4.2. On-site PUE and WUE simulation results

Figs. 5 and 6 show the simulated ranges of annual average PUE and WUE values for the 10 different DC cases and 15 different climate zones, with the lower limits corresponding to best practice efficiency values and the upper limits corresponding to poor practice efficiency values.

As expected, the simulated PUE ranges are lower in cooler climates. For almost all the cases, the highest PUE values were observed in 1A (hot moist) and the lowest values were observed in 8 (subarctic). There are three minor exceptions. In case 10, for which we assumed isothermal humidifiers for space humidification, the highest PUE value was identified in 2B (hot dry) because the required isothermal humidification gives rise to an additional DC cooling load and thus increases the PUE value. In cases 3 and 6, airside economizers (no adiabatic cooling) were used, and the lowest PUE values were predicted in 4C (marine) because of its suitability of the outdoor air temperature and humidity for direct air cooling. In cases 2 and 4, waterside economizers were used, and the lowest PUE values were associate with 3C (marine) because its dry summer season is beneficial for indirect evaporative cooling. However, these exceptions are isolated, and the general trends toward lower PUE values in cooler climates holds well.

However, the magnitudes of the PUE values differ significantly by DC size due to different efficiency practices implemented, from a median of 1.12–1.25 for large-scale DCs (case 1 and 2), to a median of 1.39–1.98 for midsize DCs (case 3–7), and to a median of 1.71–2.22 (case 8–10) for small DCs. The medians of the simulated ranges for different sizes of DCs are very similar to the values reported by LBNL (Shehabi et al., 2016), industry PUE survey data from the Uptime Institute (Lawrence, 2019), reported values by hyperscale operators (Facebook, 2020c, 2020d, 2020b, 2020a; Google, 2020), and values presented in recent academic studies (Li et al., 2020; Ni and Bai, 2017), which support the accuracy of this paper's PUE model.

Likewise, the WUE values vary by DC size due to different efficiency practices implemented but show a much greater variation across different cooling system types. For water-cooled chiller systems, DCs with no economizer use were expected to have the highest WUE values (case 5 and 8) due to significant amounts of cooling tower water use.

Table B.1
Ranges of input values for major DC facility system parameters.

Input variables	Unit	Large-scale DC		Midsize DC					Small DC			References
		Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8	Case 9	Case 10	
UPS efficiency	%	90–99	90–99	80–94	80–94	80–94	80–94	80–94	77–85	77–85	77–85	(Barroso et al., 2013; Greenberg et al., 2006; Park, 2011)
Percentage of power loss in power transformation and distribution system	%	0–2	0–2	2–5	2–5	2–5	2–5	2–5	2–4	2–4	2–4	(Barroso et al., 2013; Park, 2011; Rasmussen, 2011)
Lighting power to IT power ratio	%	0–0.2	0–0.2	2–5	2–5	2–5	2–5	2–5	2–4	2–4	2–4	(Barroso et al., 2013; Lei and Masanet, 2020; Rasmussen, 2011)
S upply air dry bulb setpoint (lower bound)	°C	10–18	10–18	15–18	15–18	15–18	15–18	15–18	18–22.5	18–22.5	18–22.5	(ASHRAE, 2015)
S upply air dry bulb setpoint (higher bound)	°C	27–35	27–35	27–32	27–32	27–32	27–32	27–32	22.5–27	22.5–27	22.5–27	(ASHRAE, 2015; Miller, 2012)
S upply air dew point setpoint (lower bound)	°C	-12 – -9	-12 – -9	-12 – -9	-12 – -9	-12 – -9	-12 – -9	-12 – -9	-9.9 – -8.1	-9.9 – -8.1	-9.9 – -8.1	(ASHRAE, 2015)
S upply air dew point setpoint (higher bound)	°C	15–27	15–27	15–27	15–27	15–27	15–27	15–27	13.5–16.5	13.5–16.5	13.5–16.5	(ASHRAE, 2015)
Supply air relative humidity setpoint (lower bound)	%	8–20	8–20	10–30	10–30	10–30	10–30	10–30	20–30	20–30	20–30	(ASHRAE, 2015)
Supply air relative humidity setpoint (higher bound)	%	60–95	60–90	60–80	60–80	60–80	60–80	60–80	54–66	54–66	54–66	(ASHRAE, 2015; Newman, 2013)
Sensible heat ratio (SHR)	%	95–99	95–99	95–99	95–99	95–99	95–99	95–99	95–99	95–99	95–99	(Evans, 2010; Power, 2010)
T emperature difference (supply/return CRAH air)	°C	13.9–19.4	13.9–19.4	5–10	5–10	5–10	5–10	5–10	5–8	5–8	5–8	(Joshi and Kumar, 2013; Lei and Masanet, 2020)
T emperature difference (supply/return facility system water)	°C	5–10	5–10	5–10	5–10	5–10	5–10	5–10	5–10	5–10	n/a	(ASHRAE, 2015,ASHRAE 2014)
T emperature difference (supply/return cooling tower water)	°C	4–6	4–6	4–6	4–6	4–6	n/a	n/a	4–6	n/a	n/a	(ASHRAE, 2014)
Fan pressure (CRAH)	Pa	300–1000	300–700	400–1000	400–900	400–900	400–1000	400–900	400–900	400–900	400–600	(Joshi and Kumar, 2013; Lei and Masanet, 2020; Patankar, 2010)
Fan efficiency (CRAH)	%	65–90	65–90	60–80	60–80	60–80	60–80	60–80	60–75	60–75	60–75	(Schild and Mysen, 2009)
Fan pressure (cooling tower)	Pa	100–400	100–400	200–400	200–400	200–400	n/a	n/a	200–400	n/a	n/a	(Lei and Masanet, 2020)
Fan efficiency (cooling tower)	%	65–90	65–90	60–80	60–80	60–80	n/a	n/a	60–75	n/a	n/a	(Schild and Mysen, 2009)
Pump pressure (humidification pump)	kPa	6300–7700	6300–7700	6300–7700	6300–7700	6300–7700	6300–7700	6300–7700	6300–7700	6300–7700	n/a	(Condair, 2020; Lei and Masanet, 2020)
Pump efficiency (humidification pump)	%	60–80	60–80	60–80	60–80	60–80	60–80	60–80	60–70	60–70	n/a	(DoE, 2006)
Pump pressure (chiller pump)	kPa	114.9–172.4	114.9–172.4	114.9–172.4	114.9–172.4	114.9–172.4	114.9–172.4	114.9–172.4	114.9–172.4	114.9–172.4	n/a	(Lei and Masanet, 2020)
Pump efficiency (chiller pump)	%	60–80	60–80	60–80	60–80	60–80	60–80	60–80	60–70	60–70	n/a	(DoE, 2006)
Pump pressure (cooling tower)	kPa	166.9–250.4	166.9–250.4	166.9–250.4	166.9–250.4	166.9–250.4	n/a	n/a	166.9–250.4	n/a	n/a	(Lei and Masanet, 2020)
Pump efficiency (cooling tower)	%	60–80	60–80	60–80	60–80	60–80	n/a	n/a	60–70	n/a	n/a	(DoE, 2006)
Pump pressure (waterside economizer pump)	kPa	n/a	114.9–172.4	n/a	114.9–172.4	n/a	n/a	n/a	n/a	n/a	n/a	(Lei and Masanet, 2020)
Pump efficiency (waterside economizer pump)	%	n/a	60–80	n/a	60–80	n/a	n/a	n/a	n/a	n/a	n/a	(DoE, 2006)
A pproach temperature (cooling tower)	°C	2.8–6.7	2.8–6.7	2.8–6.7	2.8–6.7	2.8–6.7	n/a	n/a	2.8–6.7	n/a	n/a	(ASHRAE, 2015)
A pproach temperature (economizer heat exchanger)	°C	n/a	1.7–2.8	n/a	1.7–2.8	n/a	n/a	n/a	n/a	n/a	n/a	(ASHRAE, 2015)
Chiller partial load factor	–	0.2–0.8	0.2–0.8	0.1–0.5	0.1–0.5	0.1–0.5	0.1–0.5	0.1–0.5	0.1–0.5	0.1–0.5	n/a	

(*continued on next page*)

Table B.1 (*continued*)

Liquid-gas ratio (cooling tower)	–	0.2–4	0.2–4	0.2–2	0.2–2	0.2–2	n/a	n/a	0.2–2	n/a	n/a	(Gullo et al., 2017; Lei and Masanet, 2020; Squillo, 2018; Yu and Chan, 2007)
COP relative error to regressed value	%	-11 – 11	-11 – 11	-40 – 0	-40 – 0	-40 – 0	-40 – -25	-40 – -25	-60 – -20	-45 – -30	-45 – 20	(Asvapoositkul and Treeutok, 2012; Lemouari, Boumaza and Kaabi, 2009) (Gullo et al., 2017; Lei and Masanet, 2020; Squillo, 2018; Yu and Chan, 2007)
Heat exchanger effectiveness (CRAH cooling coils)	–	n/a	0.7–0.9	n/a	0.65–0.9	0.65–0.9	0.65–0.9	0.65–0.9	0.65–0.8	0.65–0.8	n/a	(Lei and Masanet, 2020; Sammeta et al., 2011)
Windage loss of water as a percentage of cooling tower mass flow rate	%	0.005–0.5	0.005–0.5	0.005–0.5	0.005–0.5	0.005–0.5	n/a	n/a	0.05–0.5	n/a	n/a	(Environmental Protection Agency and program, 2017; Ramzy, 2020)
Cycles of concentration	–	3–15	3–15	3–12	3–12	3–12	n/a	n/a	3–12	n/a	n/a	(Environmental Protection Agency and program, 2017)

Note: uniform distributions of the input variable ranges were assumed in the uncertainty and sensitivity analysis (see section 3.4).

Within this category, the WUE values were smaller in cooler climates due to the reduced amount of heat rejected by condensers (because of higher chiller COP in cooler climates). However, the reduced heat from the chiller compressor is a small effect compared to the required steady heat rejected from the DC's IT device load. Hence, the trend of decreasing WUE values was less pronounced when moving to the cooler climates. The same reason could explain the comparatively smaller but still considerable WUE values for DCs with waterside economizers (cases 2 and 4), where the use of the economizers eliminates or reduces the heat from the compressors but the large source of internal heat from IT devices still needs to be removed by the cooling towers, and hence results in a still considerable cooling tower water consumption.

Furthermore, DCs with airside economizers (cases 1 and 3) can have much lower WUE values when outside weather conditions allow direct air cooling and the shutdown of the chilled water systems, which frequently occurs in cooler climates. Note that, although the combination of the adiabatic cooling systems with the airside economizers (case 1) incurs additional water use for humidification, it reduces the probability of running more water-intensive supplemental cooling systems. As a result, the water use of case 1 is observably lower compared to that of case 3. Moreover, the direct water use of air-cooled chillers and direct expansion systems (cases 6, 7, 9, and 10) is very small and only driven by sporadic humification needs. However, air-cooled chillers and direct expansion systems are generally less energy-efficient (in terms of PUE values) than water-cooled chiller systems. Hence, if the water embodied in the DCs' electricity use (indirect water use) was considered, such systems may have significantly higher overall water footprints than the others, which is beyond the scope of this paper.

Although the DC WUE values are climate-dependent, unlike the obvious relationship between the annual average PUE values and the climate zone as shown in Fig. 5, the annual average WUE values in Fig. 6 are less climate zone sensitive except for cases 1, 3, 5, and 8. For cases 1 and 3, the considerable difference in the whole year of available free cooling hours between the climate zones explain the changes in the associated WUE values. For cases 5 and 8, the climate effect on the annual average WUE value is still visible but to a lesser extent. This is because a large portion of the cooling tower water use is attributable to the DC's IT device heat generation, which is DC workload-dependent and climate-independent and obscures water use that is climate-dependent (i.e., the water use for rejecting the heat from the chiller compressor). Thus, this trend is solely determined the climate's effect on the chiller COP, and the decreasing trend is small given the large portion of climate-independent water use (see Eq. (4)). For cases 2 and 4, the climate effect still exists but becomes obscured due to the use of the waterside economizer, which reduces or even eliminates the climate-dependent water use part contributed by the chiller compressor. For the remaining cases 6, 7, 9 and 10, the effect of climate on the WUE values are negligible due to the small amount of DC water use.

As discussed previously, the ranges for each case and climate zone indicate the difference between best (lower values) and poor (upper values) efficiency practices, indicating the possible range of energy and water performance that can exist for any specific type of data center in a specific location. Furthermore, within any single climate zone, comparing the PUE and WUE ranges across data center cases indicates how different space conditioning technologies and practices lead to different amounts of energy and water use. As such, our ranges can help sustainability analysts make more accurate assumptions about technology- and climate-specific PUE and WUE values, inclusive of designing scenarios that represent best, poor, and average efficiency practices.

In Figs. 5 and 6 it can be seen that the ranges of PUE and WUE values are much larger for small and midsize DCs than for large-scale DCs. This is because large-scale DCs are more optimized for their systems' operation, where the use of advanced control techniques such as machine learning (Gao and Jamidar, 2014) result in less variability in their systems' efficiencies and setpoints. In contrast, small and midsize DCs may not implement such strategies, and thus have larger variations in their

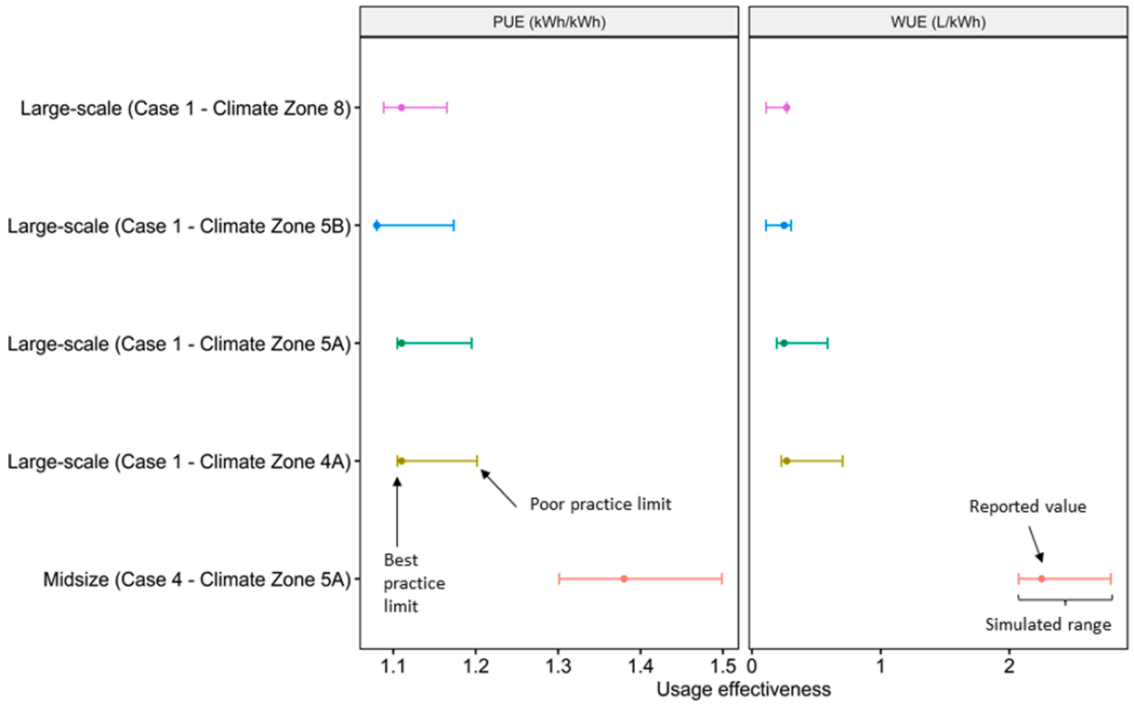


Fig. 4. Simulated PUE and WUE ranges versus reported values.

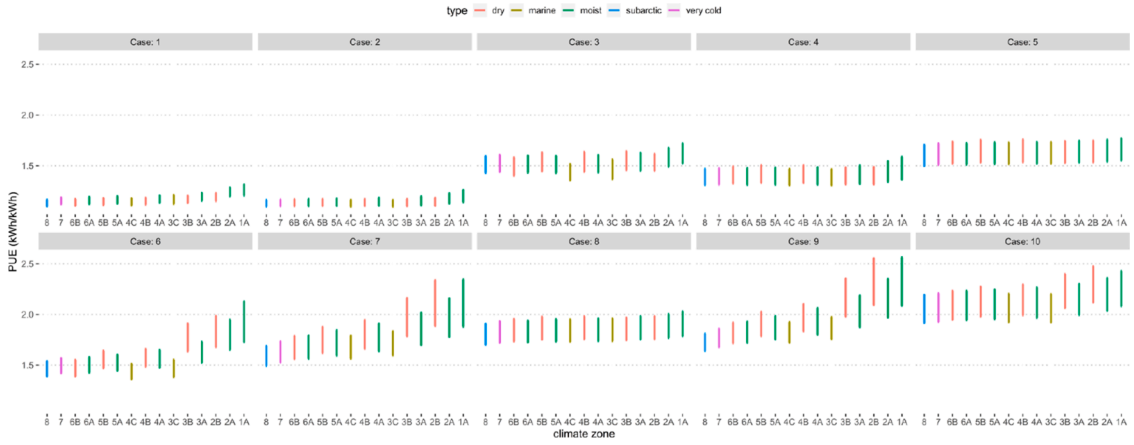


Fig. 5. Annual average PUE simulation results for 10 DC cases and 15 U.S. climate zones.



Fig. 6. Annual average WUE simulation results for 10 DC cases and 15 U.S. climate zones.

systems' operational characteristics leading to larger PUE and WUE ranges. However, compared to the ranges of PUE values, the ranges of WUE values can be much larger, especially for cases that involve cooling tower operations (cases 2, 3, 4, 5, and 8, and case 1 under hot climates). The larger WUE ranges highlight the large water saving potentials associated with efficient cooling tower operational strategies, such as choosing energy-efficient chillers to reduce the controllable part of cooling tower water evaporation, installing windage drift eliminators to reduce cooling tower windage water losses, and increasing the cooling tower water's dissolved mineral level within the acceptable range to reduce cooling tower draw-off water use (Environmental Protection Agency and program, 2017).

4.3. Climate dependent practical minimum PUE and WUE values

This section discusses the achievable practical minimum PUE and WUE values by different DC cases, represented by the lower limits of the simulation results. These practical minima can serve as useful benchmarks for sustainability analysts to understand achievable energy and water efficiencies within different climate zones, and to quantify the remaining PUE and WUE improvement potentials for DCs compared to any assumed average/typical values within the ranges.

It can be clearly observed that locations with higher latitude generally have lower practical minimum PUE and WUE values due to lower dry bulb temperatures (Fig. 3, 5, 6). The outdoor humidity level is another determining factor that affects the practical minimum PUE and WUE values, as indicated by the different minimum values that exist across the marine, dry, and moist climate zone categories. Practical minima are lower in dry climates compared to hot-humid climates, because DCs can make use of direct or indirect evaporative free cooling under dry climates. In hot-humid climates, dehumidification must occur more frequently, leading to more cooling required from mechanical chillers to process the air to the dew point.

While we acknowledge that climate is one of the most influential factors affecting the practical minimum PUE and WUE values (see section 4.4), some of the cases show very little variation (in magnitude) in PUE and WUE across climate zones (e.g., the PUEs of cases 1 and 2, and WUEs of cases 2, 4, 5, and 8). This observation conforms to the locational PUE value trends from Google (Google, 2020). However, this does not contradict with our findings in the next section because the values here correspond to the annual averages, where the averaging effect under normal and favorable weather conditions (such as those during night and winter) across the year could explain the apparent lack of climate zone variations.

Finally, achieving the practical minimum PUE and WUE values requires DCs to aggressively improve every aspect of their facility systems'



Fig. 7. Sobol's total-effect indices for the simulated DC PUE and WUE values.

efficiency, some of which could be very difficult without the help of advanced control technologies (Khalid and Wemhoff, 2019) or proper operations and expert trouble shooting. One of the most useful strategies is to focus on measures that could bring the largest saving potentials, which sustainability analysts should understand and are identified by the sensitivity analysis results discussed in the next section.

4.4. Sensitivity analysis

Fig. 7 summarizes the total-effect indices of Sobol's sensitivity analysis applied to our modeling results. A variable with a larger total-effect index has greater contribution to the variance of the model's predicted PUE or WUE values, demonstrating its importance as a factor that affects a DC's energy or water use efficiency. Similarly, a variable with a large total-effect index also indicates that it is the variable for which the sustainability analyst should try to obtain the most accurate value for more accurate PUE and WUE predictions.

It can be clearly observed that the variance in both the PUE and WUE predictions is attributable to a few key variables. Outdoor climates (dry bulb temperature and relative humidity) are one of the most common and influential variables for the predicted PUE and WUE values. This applies to nearly all the cases' PUE values and the WUE values of cases where airside economizers were used (cases 1, 3, and 6). These findings underscore the importance of using one-hour time-steps in our simulations of annual average PUE and WUE values, given that free cooling resources vary at this timescale. For DCs that mainly use cooling towers for heat removal (cases 2, 4, 5, and 8), variance in the windage water loss percentage and cycles of concentration are the largest contributors to variance in the predicted WUE values, reinforcing the importance of proper cooling tower operations for DC water efficiency. And for the cases 7, 9, and 10, the variance in the predicted WUE values is almost solely attributable to the variance in the sensible heat ratio because space humidification is the only source of water use in these cases.

When the climate conditions are given, meaning that the climate variables do not contribute to the simulation variance, the most common important variables for the DC PUE predictions become the UPS efficiency, chiller partial load factor, and a parameter describing the variabilities between different DCs' chiller efficiencies (i.e., COP relative error to regressed value (Lei and Masanet, 2020)). These findings underscore the importance of energy-efficient equipment for achieving practical minimum PUE and WUE values.

As for the WUE predictions, the most important parameters remain tthe windage water loss percentage and cycles of concentration for cases that rely on cooling towers for heat removal and, and the sensible heat ratio for cases that only implement space humidification. Furthermore, variance in the DC indoor air setpoints altogether play as one large contributor to the variance in both the PUE and WUE predictions (note that Sobol's sensitivity indices can be additive), which affect the DCs' water and energy use through influencing the amount of free cooling that can be exploited by DCs. These facility system parameters are the largest PUE and/or WUE variance contributors when the climate variables are known. On one hand, they indicate the primary factors to improve for reducing a DCs' PUE and WUE values. On the other hand, they also represent the variables most influencing modeling uncertainties in macro-level DC energy and water use estimations. These uncertainties will be challenging to reduce without empirical data on these variables being reported by existing data centers. Thus, improved data reporting from DC operators is imperative for more accurate PUE and WUE predictions by sustainability analysts.

5. Conclusions

This work addressed key barriers for sustainability analysts seeking to quantify the energy and water use of DCs, which are a rapidly growing sector. The proposed approach produced thermodynamically-compatible, best-to-poor efficiency practice PUE and WUE ranges for a wide variety of air-cooled DC archetypes in all major U.S. climate zones, whereas the accuracy of the model for its intended purposes has been validated by annual data from real DC operations. This work provides sustainability analysts with PUE and WUE values that have far greater technology and climate zone specificity than previous analyses, and enables them to consider best and worst case values for more robust DC onsite energy use and water use estimates as well as PUE and WUE improvement potentials. Moreover, the consistent PUE and WUE values presented here can facilitate more accurate modeling of indirect water use, for which PUE is the key driving variable.

Results reflect how DCs typically have lower PUE values in cooler climates, but further showed that PUE values vary greatly by DC size due to varied operating practices. Similarly, WUE values are lower in cooler climates, but show significant greater variations across different DC cooling system types. The case-by-case sensitivity analyses identified the most effective measures for PUE and WUE improvement. Results reinforced the significance of climates and indoor setpoints in determining PUE and WUE values, highlighted the importance of water-efficient operations (such as installing windage drift eliminators, or proper water dissolved mineral management) if cooling towers were used, and also underscore the necessity of deploying energy efficient equipment (including energy-efficient chillers and UPS systems). The sensitivity analysis also identified major sources of uncertainties, enabling sustainability analysts to target better data for specific variables in future analyses.

Future model accuracy also hinges on improved PUE and WUE reporting by more DCs, which would enable reduction of simulation uncertainties. Such reporting should be encouraged by policy makers to improve external understanding of DC water use but also to stimulate PUE and WUE improvements.

Although this study focused on U.S. DCs, it could be easily extended to other regions with the relevant meteorological data. The proposed modeling framework can be a valuable tool for DC multi-objective resource use optimization through energy- and water- efficient facility system management (Lei and Masanet, 2021). It could also provide granular data to further investigate the environmental benefits of DC workload mitigation/shifting (Lei et al., 2021a; Zheng et al., 2020). Additionally, this study provides a basis for constructing reduced-order models of data center energy and water use that take into account climate data and cooling system types, which could further benefit data center early-stage design and planning.

The results can also provide insights for climate-wise DC siting. However, capital investment is another important aspect correlating with the climate condition for DC site selection, which is not considered in this study and should be subject to more future research. For example, a DC with an airside economizer and adiabatic cooling could have both lower capital costs and less operational energy and water consumption in cool areas compared to hot-humid regions, due to the reduced cost for supplemental mechanical cooling system and the lowered PUE and WUE value. In addition, we only focused on WUE and PUE as means for estimating absolute direct and indirect water use quantities, respectively. Future work should consider quantities, required qualities, and local stress issues associated with DC water use.

CRediT authorship contribution statement

Nuoa Lei: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization. **Eric Masanet:** Validation, Investigation, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that cdocould have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.resconrec.2022.106323.

Appendix A. Simulation code availability

The datasets and codes associated with this study are available at: https://github.com/nuoaleon/Data-Center-Water-footprint

Appendix B. Ranges of model inputs

Table

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